

Container Orchestration Tool – Kubernetes

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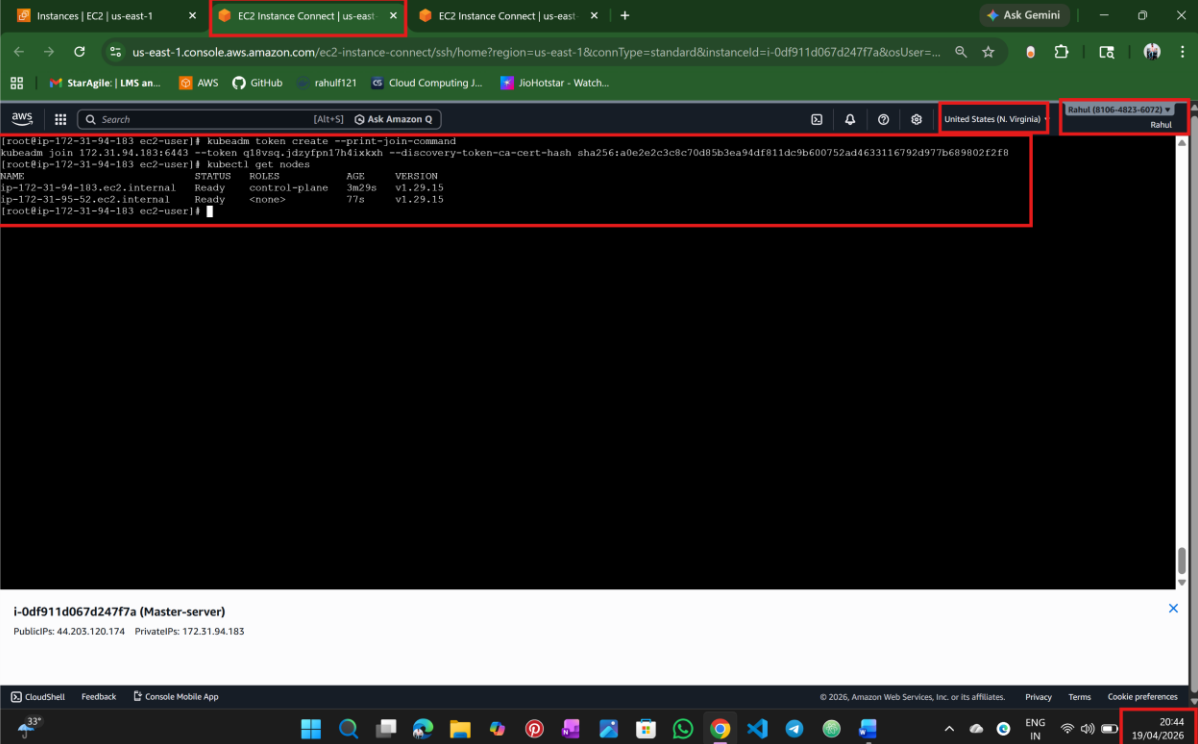
L3 - Implement Rolling-Update Strategy to Upgrade the Application Image from V1.0 to V1.1

Step:1 login into AWS console and create two ec2 instances

The screenshot displays the AWS Management Console interface for the 'us-east-1' region. The 'Instances' page is active, showing a list of two EC2 instances: 'Master-server' and 'worker-server'. Both instances are in the 'Running' state. The 'Master-server' has an Instance ID of 'i-0d9f91d067d24777a' and is of type 'c7i-flex.large'. The 'worker-server' has an Instance ID of 'i-02bfbcb07a60fd1f' and is also of type 'c7i-flex.large'. The console shows various monitoring metrics like CPU utilization, network in/out, and network packets in/out, all of which currently show 'No data available'. The top navigation bar includes the AWS logo, search bar, and user profile 'Rahul (8106-4823-6072)'. The bottom status bar shows the date and time as '19/04/2026 20:36'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4
Master-server	i-0d9f91d067d24777a	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b	ec2-44-203-120-174.co...	44.203.120.174
worker-server	i-02bfbcb07a60fd1f	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b	ec2-3-88-40-15.comput...	3.88.40.15

Step:2 This is Master nose terminal



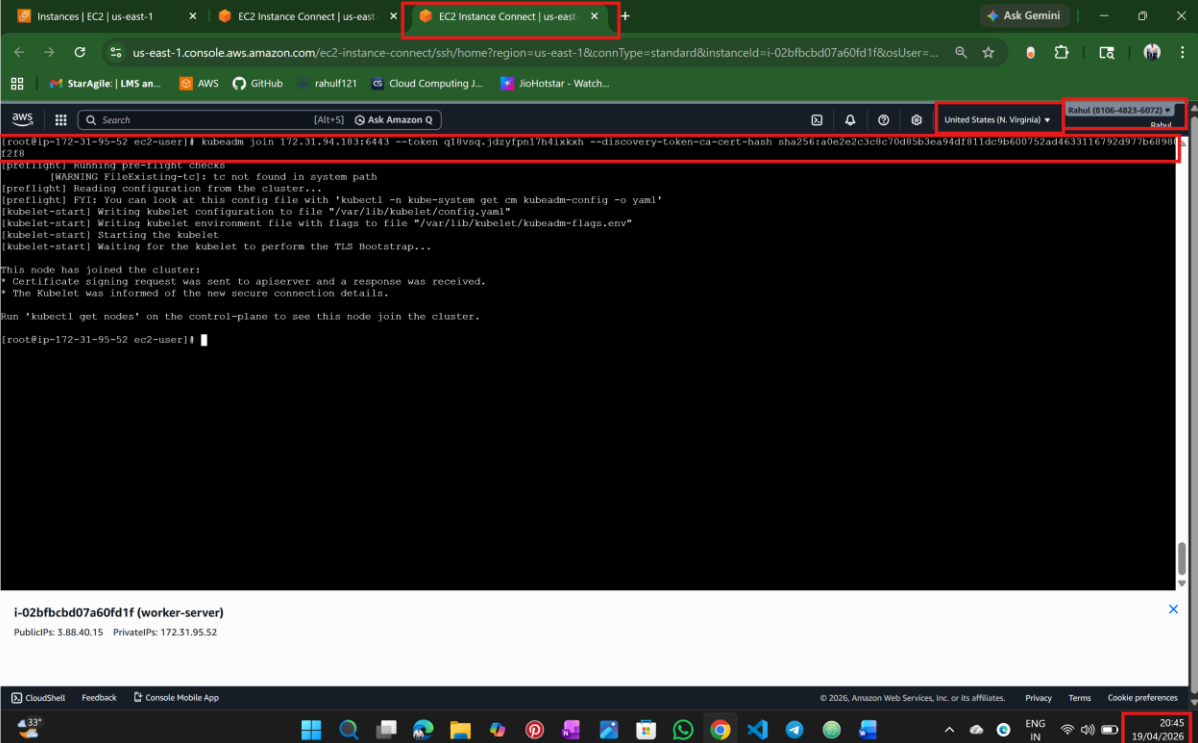
The screenshot shows the AWS console terminal for the Master node (i-0df911d067d247f7a). The terminal window is titled "EC2 Instance Connect | us-east-1". The terminal output shows the following commands and results:

```
[root@ip-172-31-94-183 ec2-user]# kubeadm token create --print-join-command
kubeadm join 172.31.94.183:6443 --token q18vsg.jdzyfpl7h4ixkxh --discovery-token-ca-cert-hash sha256:a0e2e2c3c8c70d85b3ea94df811dc9b600752ad4633116792d977b689802f2f8
[root@ip-172-31-94-183 ec2-user]# kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
ip-172-31-94-183.ec2.internal        Ready    control-plane   3m29s   v1.29.15
ip-172-31-95-52.ec2.internal        Ready    <none>         77s     v1.29.15
[root@ip-172-31-94-183 ec2-user]#
```

The terminal output also shows the node details for the Master node (i-0df911d067d247f7a):

```
PublicIP: 44.203.120.174 PrivateIP: 172.31.94.183
```

Step:3 This is worker node Terminal



The screenshot shows the AWS console terminal for the Worker node (i-02bfbcb07a60fd1f). The terminal window is titled "EC2 Instance Connect | us-east-1". The terminal output shows the following commands and results:

```
[root@ip-172-31-95-52 ec2-user]# kubeadm join 172.31.94.183:6443 --token q18vsg.jdzyfpl7h4ixkxh --discovery-token-ca-cert-hash sha256:a0e2e2c3c8c70d85b3ea94df811dc9b600752ad4633116792d977b689802f2f8
[preflight] Running pre-flight checks
[WARNING FileExisting-tc]: tc not found in system path
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap...

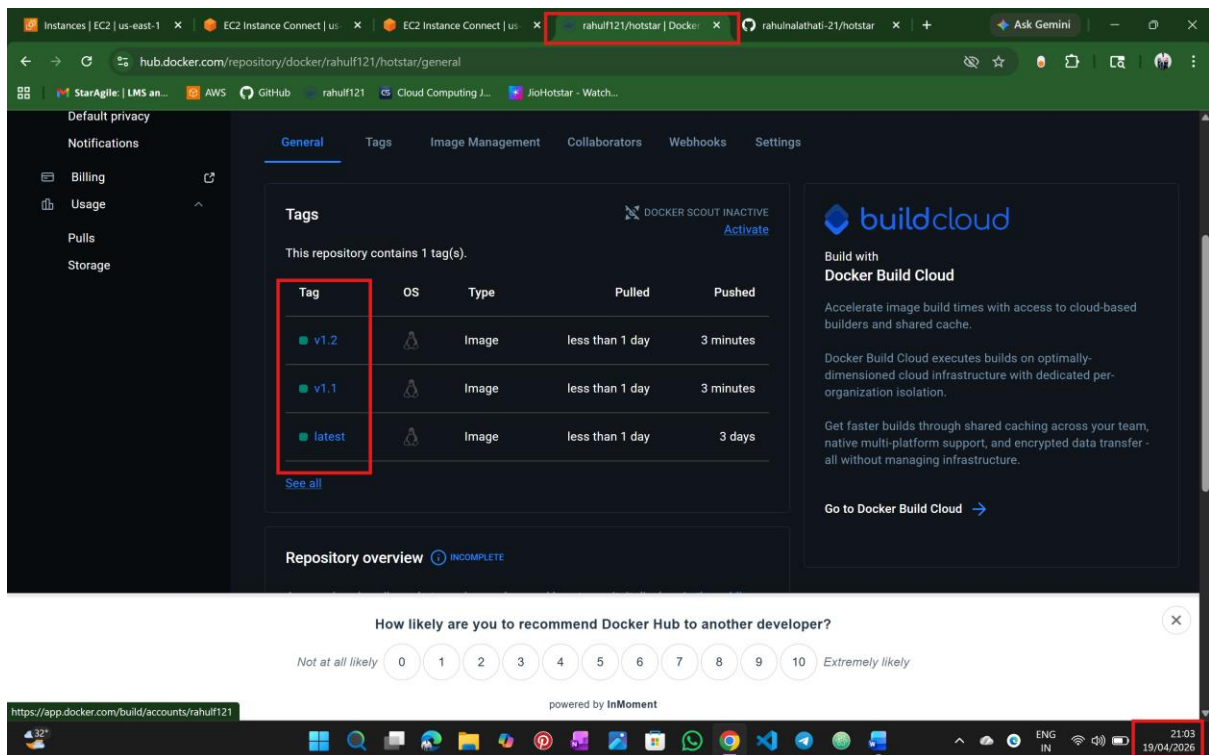
This node has joined the cluster:
* Certificate signing request was sent to apiserer and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.
[root@ip-172-31-95-52 ec2-user]#
```

The terminal output also shows the node details for the Worker node (i-02bfbcb07a60fd1f):

```
PublicIP: 3.88.40.15 PrivateIP: 172.31.95.52
```

Step:4 I already have various version of the docker images

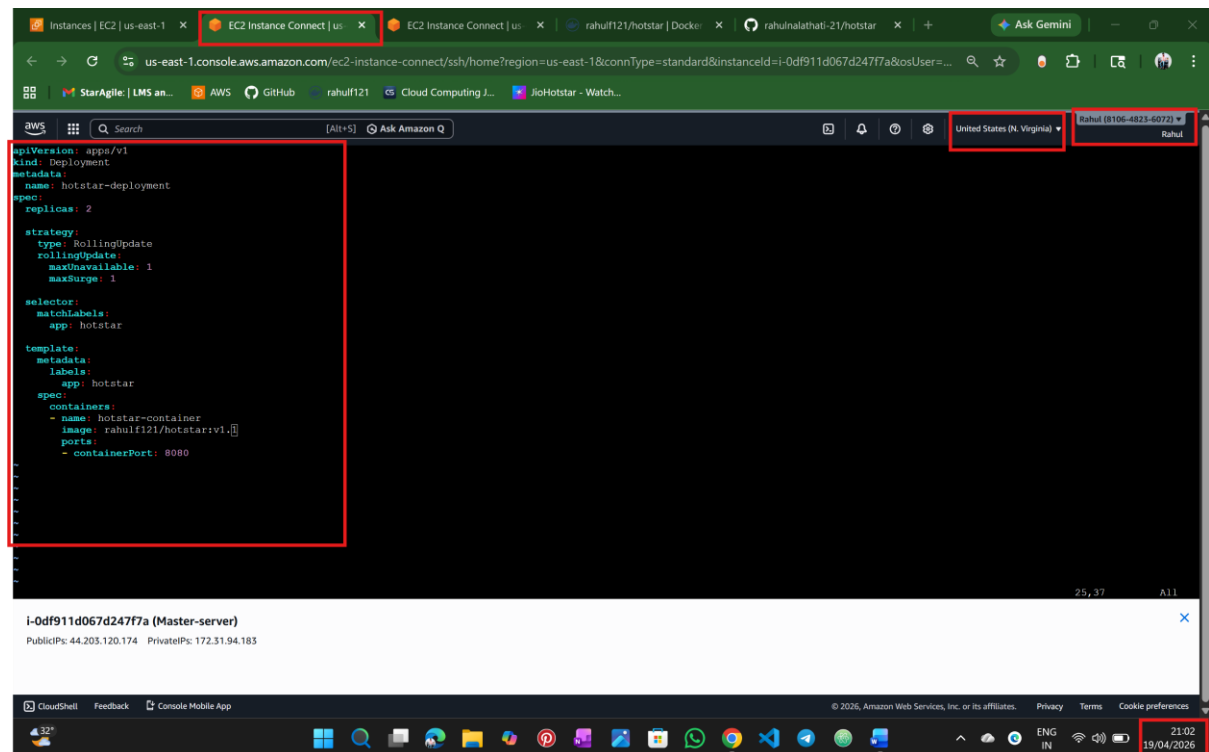


The screenshot shows the Docker Hub repository page for `rahulf121/hotstar`. The browser tabs include "Instances | EC2 | us-east-1", "EC2 Instance Connect | us...", "rahulf121/hotstar | Docker", and "rahulnalathati-21/hotstar". The URL is `hub.docker.com/repository/docker/rahulf121/hotstar/general`. The left sidebar shows navigation options like "Default privacy", "Notifications", "Billing", "Usage", "Pulls", and "Storage". The main content area has tabs for "General", "Tags", "Image Management", "Collaborators", "Webhooks", and "Settings". The "Tags" tab is active, showing a table of tags. A red box highlights the "Tag" column in the table.

Tag	OS	Type	Pulled	Pushed
v1.2	linux	Image	less than 1 day	3 minutes
v1.1	linux	Image	less than 1 day	3 minutes
latest	linux	Image	less than 1 day	3 days

Below the table is a "Repository overview" section with an "INCOMPLETE" status. To the right is a "buildcloud" advertisement. At the bottom, there is a survey question: "How likely are you to recommend Docker Hub to another developer?" with a rating scale from 0 to 10. The browser's taskbar at the bottom shows the time as 21:03 on 19/04/2026.

Step:5 This is the deployment.yaml for v1.1 version

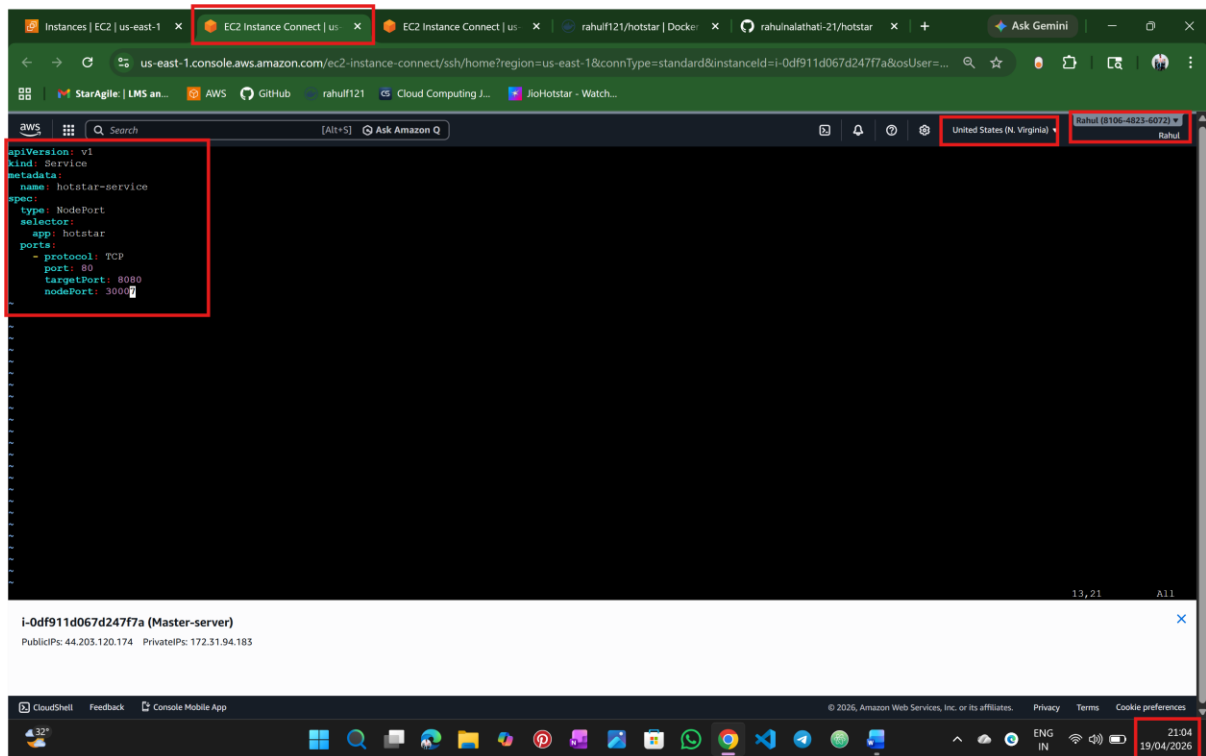


The screenshot shows the AWS Management Console for the "us-east-1" region. The browser tabs include "Instances | EC2 | us-east-1", "EC2 Instance Connect | us...", "rahulf121/hotstar | Docker", and "rahulnalathati-21/hotstar". The URL is `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0df911d067d247f7a&osUser=...`. The left sidebar shows the "aws" logo and a search bar. The main content area displays a deployment.yaml file. A red box highlights the entire content of the file.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hotstar-deployment
spec:
  replicas: 2
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  selector:
    matchLabels:
      app: hotstar
  template:
    metadata:
      labels:
        app: hotstar
    spec:
      containers:
        - name: hotstar-container
          image: rahulf121/hotstar:v1.1
          ports:
            - containerPort: 8080
```

Below the file content, there is a notification for the instance `i-0df911d067d247f7a (Master-server)` with public and private IP addresses. The browser's taskbar at the bottom shows the time as 21:02 on 19/04/2026.

Step:6 This is service.yaml

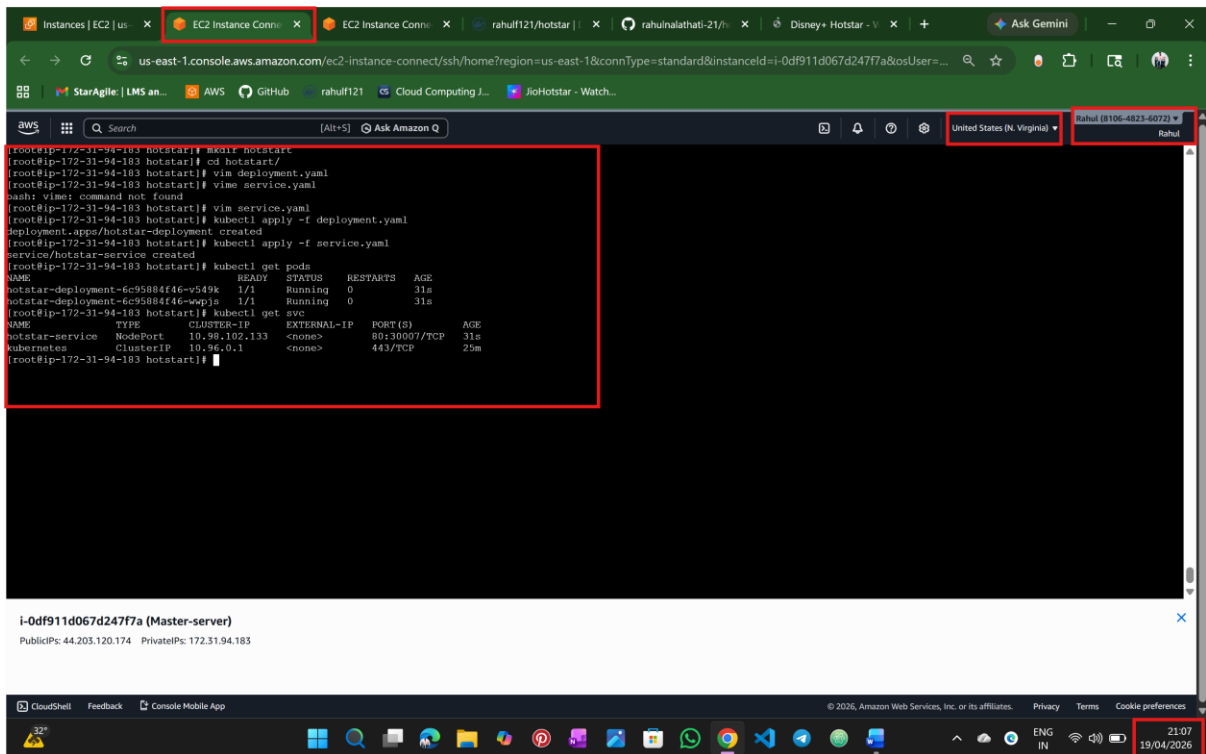


The screenshot shows the AWS Management Console interface. The top navigation bar includes tabs for 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us...', and 'EC2 Instance Connect | us...'. The main content area displays the 'service.yaml' file in a code editor. The file content is as follows:

```
apiVersion: v1
kind: Service
metadata:
  name: hotstar-service
spec:
  type: NodePort
  selector:
    app: hotstar
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30007
```

The bottom status bar shows the instance 'i-0df911d067d247f7a (Master-server)' with public IP 44.203.120.174 and private IP 172.31.94.183. The bottom right corner of the console shows the time '21:04' and date '19/04/2026'.

Step:7 apply and verify

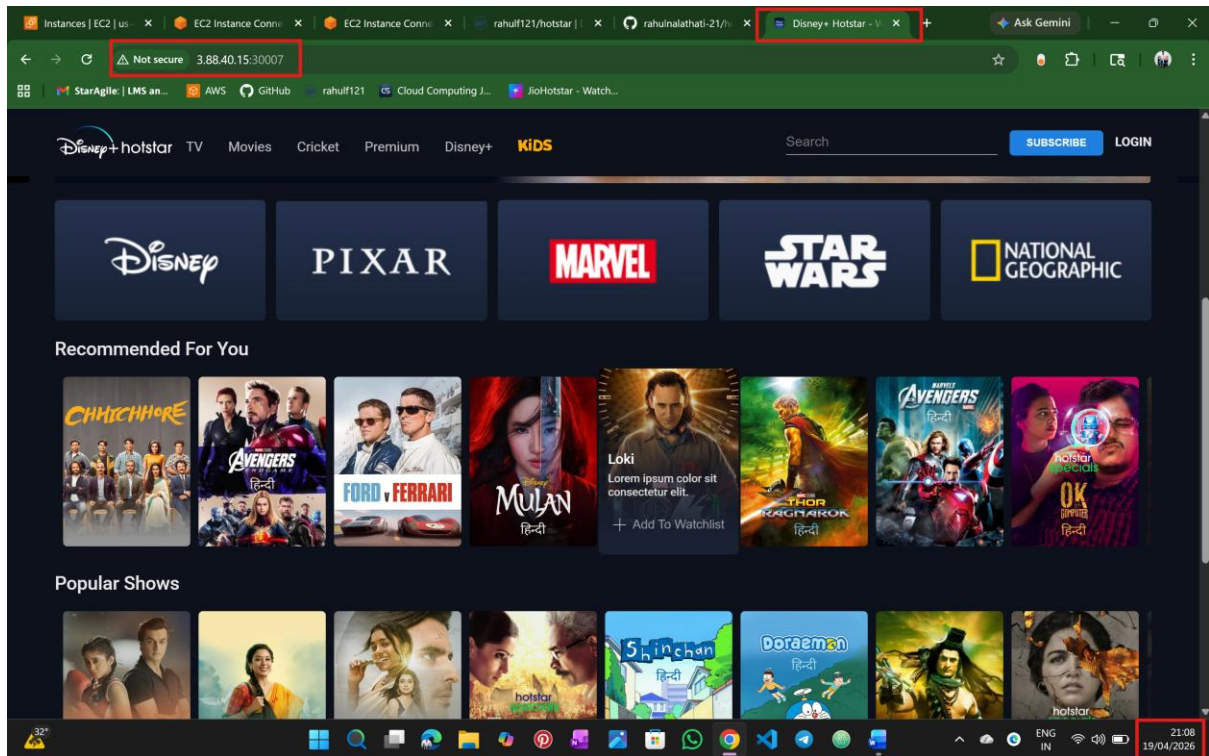


The screenshot shows the AWS Management Console interface. The top navigation bar includes tabs for 'Instances | EC2 | us...', 'EC2 Instance Connect | us...', and 'EC2 Instance Connect | us...'. The main content area displays the terminal output for applying and verifying the service. The output is as follows:

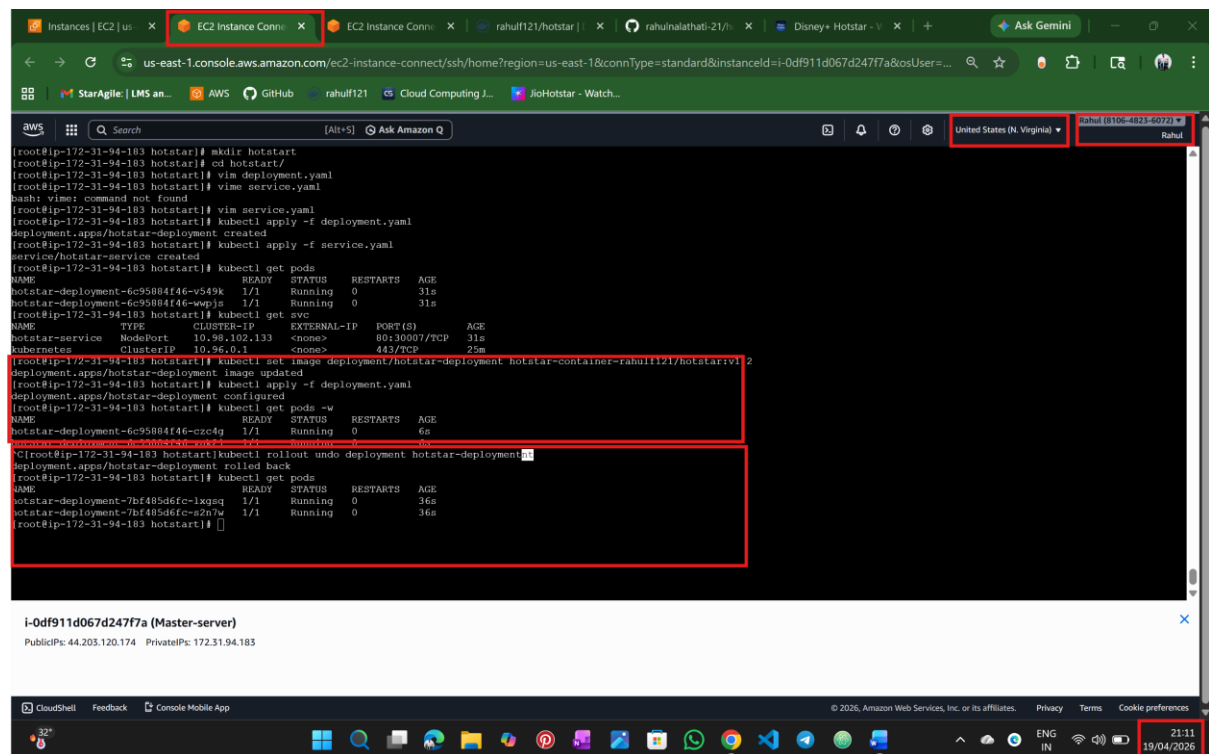
```
[root@ip-172-31-94-183 hotstar]# kubectl apply -f deployment.yaml
deployment.apps/hotstar-deployment created
[root@ip-172-31-94-183 hotstar]# kubectl apply -f service.yaml
service/hotstar-service created
[root@ip-172-31-94-183 hotstar]# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hotstar-deployment-6c95884f46-v549k 1/1     Running   0           31s
hotstar-deployment-6c95884f46-wvpjs 1/1     Running   0           31s
[root@ip-172-31-94-183 hotstar]# kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
hotstar-service NodePort    10.98.102.133 <none>        80:30007/TCP    31s
kubernetes ClusterIP   10.96.0.1     <none>        443/TCP         25m
```

The bottom status bar shows the instance 'i-0df911d067d247f7a (Master-server)' with public IP 44.203.120.174 and private IP 172.31.94.183. The bottom right corner of the console shows the time '21:07' and date '19/04/2026'.

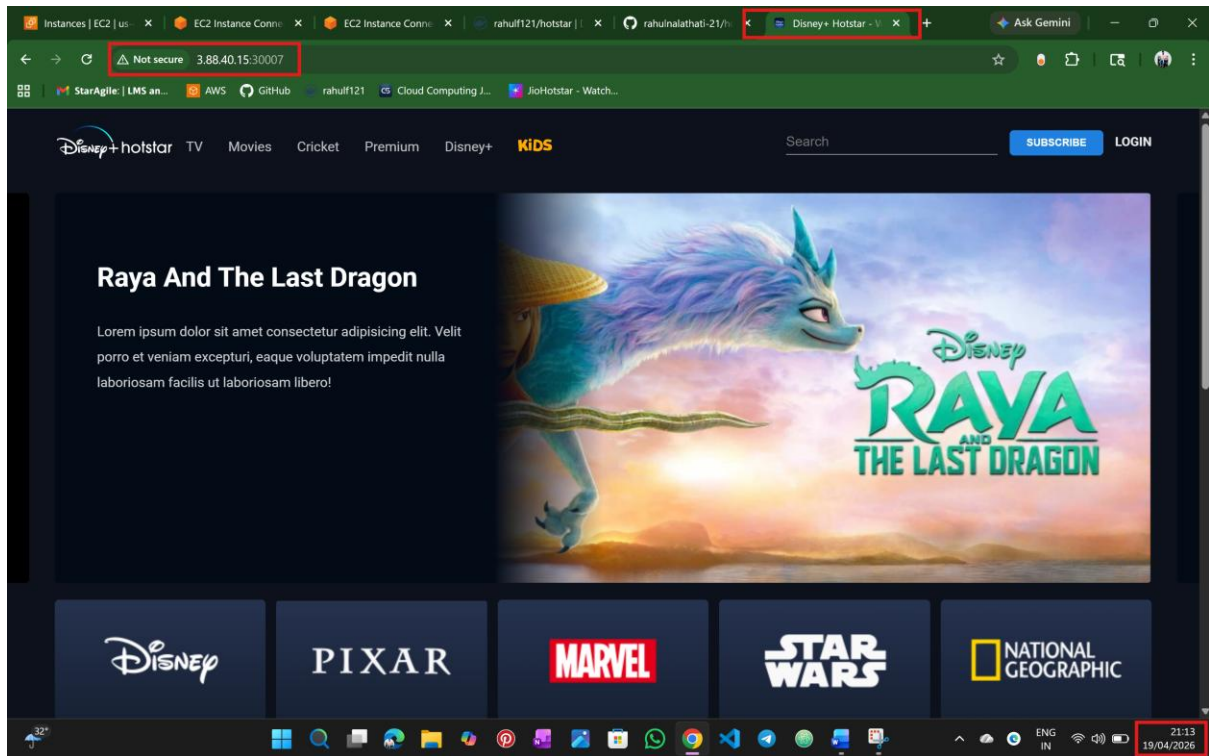
Step:8 access the application via version v1.1



Step:9 update to new version v1.2 and Rollback to the previous version v1.1



Step:10 still application is running and there is no down time



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Nalathati Rahul